

Formal Verification of Quantum Codes with Transversal Diagonal Gates in Lean 4

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arXiv:2510.20728 · github.com/LionSR/lean-qec

Transversal Gates & the SSLP Pipeline

Distance-2 Results

Lean 4 Formalization

Distance-3 Extension

Outlook

Summary

This Talk

Starting point: arXiv:2510.20728 catalogued 14,116 distance-2 codes with transversal diagonal gates via the SSLP pipeline.

Three new contributions:

1. **Lean 4 formalization** of the full SSLP pipeline — machine-checked proofs, zero sorry
2. **Analytical families:** one proof covers *infinitely many* codes
3. **Distance-3 generalization:** exact solutions for $((7, 2, 3))$ codes discovered by GPT-5.2, formally verified in Lean



Transversal Gates & the SSLP Pipeline

Transversal Diagonal Gates

Physical gate with angle vector $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{Z}_m^n$:

$$U(\mathbf{a}, m) = \bigotimes_{i=1}^n \begin{pmatrix} 1 & 0 \\ 0 & \omega_m^{a_i} \end{pmatrix}, \quad \omega_m = e^{2\pi i/m}$$

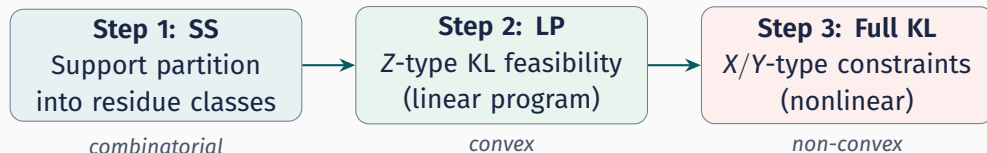
Action on computational basis: $U(\mathbf{a}, m) |x\rangle = \omega_m^{\langle \mathbf{a}, x \rangle} |x\rangle$ where $\langle \mathbf{a}, x \rangle = \sum_i a_i x_i$.

Define **residue classes** $C_b(\mathbf{a}) = \{x \in \{0, 1\}^n : \sum_i a_i x_i \equiv b \pmod{m}\}$.

Eigenstate condition

$$\text{supp}(|j_L\rangle) \subseteq C_{S_j}(\mathbf{a}) \implies U(\mathbf{a}, m) |j_L\rangle = \omega_m^{S_j} |j_L\rangle$$

Subset-Sum Linear Programming: systematic code construction.



Step 2 is an LP $\Rightarrow$ fast rejection of infeasible parameters.

For **distance 2**: our residue-shift screen makes Step 3 *automatic*.

For **distance 3**: we solve Step 3 via exact arithmetic in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, i)$.

Step 1: Subset-Sum Support Partition

Residue classes $C_b(\mathbf{a}) = \{x \in \{0,1\}^n : \sum_i a_i x_i \equiv b \pmod{m}\}$ partition $\{0,1\}^n$.

- **Disjoint:** $C_b \cap C_{b'} = \emptyset$ for $b \neq b' \Rightarrow$ automatic orthogonality
- **Logical action:** $U |j_L\rangle = \omega_m^{S_j} |j_L\rangle$
- **Projective order:**
 $\mathcal{O} = m / \gcd(m, S_1 - S_0, \dots)$

Example: $n=5, m=7$

$\mathbf{a}=(1, 1, 2, 2, 2), (S_0, S_1) = (0, 4)$

$C_0 = \{00000, 01111, 10111\}$

$C_4 = \{00011, 00101, \dots\}$ (6 elts)

$\mathcal{O} = 7 / \gcd(7, 4) = 7$

Step 2: Z-Type KL as Linear Program

Probabilities $p_{j,x} = |c_{j,x}|^2$ on support of $|j_L\rangle$.

Z-type KL: $\langle j_L | Z_i | j_L \rangle = t_i$ independent of j becomes

$$\sum_{x \in C_{S_j}} (-1)^{x_i} p_{j,x} = t_i \quad \forall i \in [n], \forall j$$

Linear in $p_{j,x}$ with constraints:

- $\sum_{x \in C_{S_j}} p_{j,x} = 1$
- $p_{j,x} \geq 0$

Standard LP feasibility.

Geometric view: sign vectors

$v(x) = ((-1)^{x_1}, \dots, (-1)^{x_n})$ are vertices.

Feasibility = nonempty $\bigcap_j \text{conv}(V_j)$.

Residue-Shift Screen: Distance 2 for Free

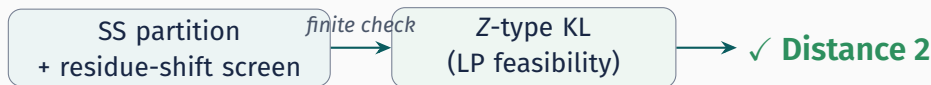
Flipping bit i shifts the residue by $\pm a_i$:

$$\langle \mathbf{a}, \mathbf{x} \oplus \mathbf{e}_i \rangle \equiv \langle \mathbf{a}, \mathbf{x} \rangle \pm a_i \pmod{m}$$

Residue-Shift Screen

$S_j - S_k \not\equiv \pm a_i \pmod{m}$ for all $i \in [n]$, all $j \neq k \implies$ union support has $d_H \geq 2$.

When the screen passes, **all X/Y-type KL vanish automatically.**



Binary Dihedral Symmetry ($K=2$)

Binary dihedral group BD_{2m} :

- Generated by $Z(2\pi/m)$ and $\hat{X} = -iX$
- $|BD_{2m}| = 4m$; $m=8 \Rightarrow$ transversal T
- Transversal $\bar{X} = X^{\otimes n}$: $|1_L\rangle = X^{\otimes n} |0_L\rangle$
- Support: $S_1 = \bar{S}_0$ (bitwise complements)
- Amplitudes: $c_{1,\bar{x}} = c_{0,x}$

BD angle condition

$$\sum_{j=1}^n a_j \equiv -1 \pmod{m}$$

Z-balance

Complement symmetry forces

$$\sum_{x \in S_0} (-1)^{x_i} p_x = 0 \text{ for all } i.$$

Only $|0_L\rangle$ needs specification.

Distance-2 Results

Systematic search via SSLP:

- $K \in \{2, 3, 4\}$, $n \leq 6$, $m \leq 18$
- Residue-shift screen for $d=2$
- LP solver + rational reconstruction

14,116

certified $((n, K, 2))$ codes

Highlights

- Orders $\mathcal{O} = 2$ through 18 at $n = 6$
- Mostly **nonadditive**
- Includes $K = 3, 4$ codes

Analytical families

Family I: $C_0 = \{0^n, 1^n\}$,
closed-form $p(0^n) = 1 - s/m$
✓ verified in Lean for all (n, m, s)

Analytical Families: One Proof, Infinitely Many Codes

Find a **closed-form pattern**, prove it *once*, instantiate everywhere.

Family I: the simplest codes

Support $C_0 = \{0^n, 1^n\}$ — just two basis states.

$$|0_L\rangle = \sqrt{p} |0 \cdots 0\rangle + \sqrt{1-p} |1 \cdots 1\rangle$$

Z-balance forces p **uniquely**:

$$p = 1 - s/m.$$

Valid for *all* $(n, m, \mathbf{a})$ passing the screen.

λ -families

Larger supports, free parameter λ :

- λ -Family II: $|C_0| = 4$
- λ -Family V: $|C_0| = 6$
- 5 families, each a **universally quantified** Lean theorem

One Lean proof $\Leftrightarrow$
infinitely many
verified codes

Worked Example: $((5, 2, 2))$ Code

$n=5, \mathbf{a}=(1, 1, 2, 2, 2), m=7, (S_0, S_1) = (0, 4)$

SS: $C_0 = \{00000, 01111, 10111\}$, C_4 : 6 elements

Screen: $4 \not\equiv \pm 1, \pm 2 \pmod{7}$ ✓

LP: $p_{00000} = 3/7, p_{01111} = p_{10111} = 2/7$
 $\langle Z_i \rangle = 3/7$ ($i=1, 2$), $-1/7$ ($i=3, 4, 5$)

Lean: one-line proof

```
def myCode : ExampleData where
  n := 5; m := 7; K := 2
  a := ![1, 1, 2, 2, 2]
  S := ![0, 4]
  supp := ![supp0, supp1]
  prob := ![prob0, prob1]

theorem verified : myCode.OK :=
  by native_decide
```

SS + screen + LP $\Rightarrow$ certified $((5, 2, 2))$ code, order $\mathcal{O} = 7$. Build: < 1 s.

Lean 4 Formalization

What is Lean 4?

A **proof assistant** based on dependent type theory:

- Every proof is a program; the **kernel** type-checks it
- Compiles $\Rightarrow$ proof is correct. No trust required.
- **Mathlib**: 180K+ formalized theorems

Key tactic: `native_decide`

- Proposition is `Decidable`? Compile to native code $\rightarrow$ run $\rightarrow$ true
- Kernel accepts: proof complete
- “All 210 KL constraints hold?” $\rightarrow$ true $\rightarrow$ QED

Landmark projects

- Freiman–Ruzsa conj. (Tao)
- Liquid Tensor Experiment
- Sphere eversion
- **This work**: quantum codes

Why Lean for QEC?

14K codes: too many for humans.

AI-found solutions: trust but verify.

Formal proof = **certainty**.

Lean 4 Codebase: lean-qec

Core Library (ss/)

BitString.lean
Basic.lean
DiagonalAction.lean
ResidueShiftScreen.lean
ZTypeKL.lean
BDSymmetry.lean
Verify.lean
FamilyI/ (6 files)
LambdaV2/ (5 families)

Distance-3 Modules

Pauli.lean
FullKL.lean
QSqrt23i.lean
FullKLEval.lean
Distance3Verify.lean
BD16Defs.lean

Codebase

~4,300 lines, ~30
modules
0 sorry, 0 warnings

Examples

Examples/D2/
Ex522.lean ((5,2,2))
Ex622.lean ((6,2,2))
Examples/D3/
BD16v1.lean through
BD16v6a.lean (8 codes)

Build

3,091 jobs
0 errors · 0 warnings

Formalization: Key Theorems

Dact_eq_global_of_SS

$$\text{supp}(|j_L\rangle) \subseteq C_{S_j} \Rightarrow U |j_L\rangle = \omega_m^{S_j} |j_L\rangle$$

no_hamming1_neighbor_of_screen

Screen condition $\Rightarrow d_H(x, y) \geq 2$
across classes
 $\Rightarrow$ all X/Y-type KL vanish

ExampleData.OK

Bundles four checks:

1. SS support condition
2. Residue-shift screen
3. Probability normalization
4. Z-type KL matching

All Decidable $\Rightarrow$
native_decide in $< 1s$

Technique: BitString $n = \text{Fin } n \rightarrow \text{Bool}$ has 2^n elements. All quantifiers over finite support are decidable $\Rightarrow$ Lean compiles to native code, executes, returns true $\Rightarrow$ kernel accepts as proof.

Distance-3 Extension

The Distance-3 Barrier

Why was distance 2 “easy”?

- Residue-shift screen kills all X/Y-type KL automatically
- Only Z-type survives $\Rightarrow$ linear in probabilities $p_x \geq 0$
- **No complex amplitudes needed**

Why is distance 3 harder?

- Must check all weight- ≤ 2 Paulis: $Z_i Z_j, X_i X_j, X_i Y_j, \dots$
- X, Y **flip bits** $\Rightarrow$ couples different basis states
- KL becomes **quadratic** in $c_x \in \mathbb{C}$
- **Relative phases matter**

The gap: distance-2 = convex optimization over $\mathbb{R}_{\geq 0}$.

Distance-3 = nonlinear system over $\mathbb{C}$. At $n=7$: 210 weight- ≤ 2 Pauli constraints.

Core challenge: solve 210 nonlinear constraints over $\mathbb{C}$, then verify with mathematical certainty.

Distance 3: Three Key Ideas

1. Most constraints vanish.

$$S_0 \cap (S_0 \oplus z_p) = \emptyset$$

$\Rightarrow$ sum is empty $\Rightarrow 0$.

BD16v1: 25 / 210

> 88% are automatic.

2. Exact number field.

Amplitudes in

$$\mathbb{Q}(\sqrt{2}, \sqrt{3}, i)$$

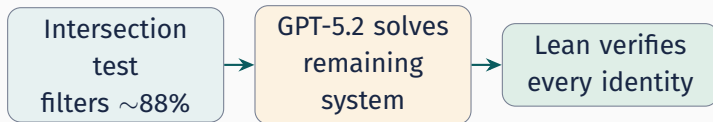
= 8 rational coefficients.

DecidableEq, no floats.

3. AI + Lean.

GPT-5.2 solves the surviving constraints symbolically.

Lean verifies *exactly*.



Why Most KL Constraints Vanish

$\langle 0_L | P | 0_L \rangle$ sums over pairs where x and $x \oplus z_P$ are *both* in S_0 .



Disjoint $\Rightarrow$ empty sum $\Rightarrow$ automatically 0

$$S_0 \cap (S_0 \oplus z_P) = \emptyset \implies \langle 0_L | P | 0_L \rangle = 0.$$

Why so effective

S_0 is **sparse**: $|S_0| = 7$ out of $2^7 = 128$. Random shifts almost never land back inside S_0 . Only 25 / 210 survive (BD16v1).

In Lean: `klIsAutomatic` is Decidable — checked by `native_decide`.

From Constraints to Solutions: GPT-5.2

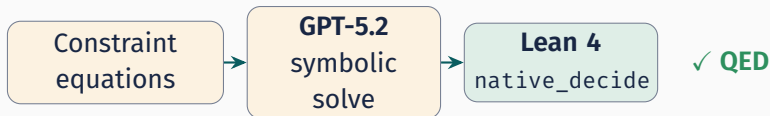
After intersection test, the surviving constraints form a **quadratic system** in $c_x \in \mathbb{C}$.

The problem:

- Unknowns: c_x for each $x \in S_0$
- $\sum_x \bar{c}_x c_{x'} \cdot (\text{phase}) = 0$
- Non-convex $\Rightarrow$ hard for standard solvers

GPT-5.2's approach:

- Given S_0 , probabilities, constraint eqns
- Exploits symmetries, groups terms
- Returns **closed-form** amplitudes in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, i)$



Exact Arithmetic in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, i)$

$m=8 \Rightarrow \omega_8 = \frac{1+i}{\sqrt{2}}$. Amplitudes involve $\sqrt{2}, \sqrt{3}$ from normalization.

Representation (qsqrt23i)

Each element $= (q_0, \dots, q_7) \in \mathbb{Q}^8$:

$$(q_0 + q_1\sqrt{2} + q_2\sqrt{3} + q_3\sqrt{6}) + (\dots)i$$

Multiplication closed.

Equality *decidable* (rational coefficients).

No floating point. Every operation exact over $\mathbb{Q}$. DecidableEq $\Rightarrow$ native_decide verifies each KL identity.

BD16v1 amplitudes (GPT-5.2)

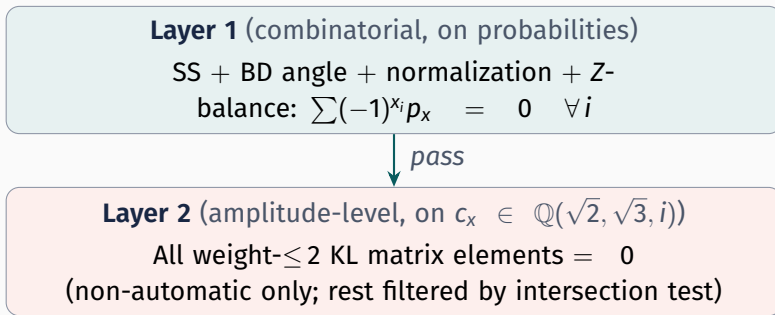
$$c_0 = \frac{1}{4} \quad c_5 = \frac{1}{4}$$

$$c_1 = c_2 = \frac{\sqrt{3}}{4} \quad c_6 = \frac{i}{2}$$

$$c_3 = c_4 = \frac{\sqrt{2}}{4}$$

In Lean: $c_1 = (0, 0, \frac{1}{4}, 0, 0, 0, 0, 0)$ ✓

Two-Layer Verification Architecture



Both layers are **fully decidable** — verified by `native_decide`.

No sorry, no axioms. Build time per code: 2–11 s.

BD16 Example: $((7, 2, 3))$ with $\mathbf{a} = (1, 2, 2, 2, 2, 3, 3)$

$m=8$ (transversal T), $|S_0| = 7$.

$$4|0_L\rangle = |0000000\rangle + \sqrt{3}|0111100\rangle + \sqrt{3}|1001110\rangle + \sqrt{2}|1010101\rangle + \sqrt{2}|1011001\rangle + |1110010\rangle + 2i|0101010\rangle$$

```
def bd16Data : Distance3Data 7 8 where
  a := ![1, 2, 2, 2, 2, 3, 3].
  b := 0.
  supp := {s0, s1, s2, s3, s4, s5, s6}.
  prob := ... -- (1,3,3,2,2,1,4)/16

theorem layer1 : bd16Data.Layer10K
  := by native_decide
theorem layer2 : AllKLSatisfied _supp _amp 2
  := by native_decide
```

Check	Result
SS + BD + norm	✓ native_decide
Z-balance	✓ native_decide
Non-auto KL	25 / 210
All 25 = 0	✓ native_decide

Complete $d \geq 3$ proof

Build: 2.6 s

8 Verified BD16 ((7, 2, 3)) Codes in Lean

All codes: $n=7$, $K=2$, $m=8$ (transversal T), verified `Layer10K + AllKLSatisfied`.

File	Angle vector $\mathbf{a}$	$ S_0 $	λ^*	Solution type	Status
BD16v1	(1, 2, 2, 2, 2, 3, 3)	7	$\sqrt{21}/8$	2-param continuous	✓ <code>native_decide</code>
BD16v2	(1, 1, 2, 2, 2, 2, 5)	10	$\sqrt{33}/4$	192-elt discrete	✓ <code>native_decide</code>
BD16v3a	(0, 1, 2, 2, 3, 3, 4)	7	$\sqrt{51}/4$	Solution A (of 3)	✓ <code>native_decide</code>
BD16v3b	(0, 1, 2, 2, 3, 3, 4)	8	$\sqrt{23}/4$	Solution B (of 3)	✓ <code>native_decide</code>
BD16v3c	(0, 1, 2, 2, 3, 3, 4)	8	$\sqrt{30}/4$	Solution C (of 3)	✓ <code>native_decide</code>
BD16v4	(1, 1, 1, 1, 3, 3, 5)	13	$\sqrt{177}/8$	continuous	✓ <code>native_decide</code>
BD16v5	(1, 1, 1, 1, 3, 4, 4)	12	$\sqrt{62}/4$	128-elt discrete	✓ <code>native_decide</code>
BD16v6a	(1, 1, 1, 2, 2, 4, 4)	16	1	uniform magnitude	✓ <code>native_decide</code>

Remaining: 3 vectors need $\sqrt{5}/\sqrt{10}$ field extension; 2 proven impossible. 12 vectors total
 $\Rightarrow$ 10 feasible, 6 verified in Lean (8 distinct codes).

What the Surviving Constraints Look Like

After the intersection test, ~ 25 – 150 non-automatic Pauli strings remain.

Diagonal ($Z_i Z_j$)

$$\langle 0_L | Z_i Z_j | 0_L \rangle = \sum_{x \in S_0} (-1)^{x_i + x_j} |c_x|^2$$

Linear in probabilities $p_x = |c_x|^2$.

Already verified in **Layer 1**.

Key simplification

BD symmetry $|1_L\rangle = X^{\otimes n} |0_L\rangle$

$\Rightarrow$ off-diagonal eqns **pair up**:

$$\langle 0 | P | 0 \rangle = 0 \Leftrightarrow \langle 1 | P | 1 \rangle = 0$$

Only $|0_L\rangle$ amplitudes needed.

Off-diagonal ($X_i X_j, X_i Y_j, \dots$)

$$\langle 0_L | P | 0_L \rangle = \sum_{\substack{x \in S_0 \\ x \oplus z_P \in S_0}} \bar{c}_x c_{x \oplus z_P} \omega^{f(x, P)}$$

Quadratic in amplitudes $c_x \in \mathbb{C}$.

Phases $\omega^{f(x, P)}$ depend on x_P bits.

BD16v1: 7 nonzero amplitudes, 25 non-automatic constraints. Probabilities fixed by Layer 1 $\rightarrow$ only phases remain.

The BD_{16} Solution Landscape

All 12 SS-LP-passing BD_{16} $((7, 2, 3))$ angle vectors:

Angle vector $\mathbf{a}$	$ S_0 $	Non-auto	Solution type	Status
(1, 2, 2, 2, 2, 3, 3)	18	83	2-param continuous	Lean ✓
(1, 1, 2, 2, 2, 2, 5)	14	97	192-elt discrete	Lean ✓
(0, 1, 2, 2, 3, 3, 4)	16	68	3 isolated solutions	Lean ✓
(1, 1, 1, 1, 3, 3, 5)	13	81	continuous	Lean ✓
(1, 1, 1, 1, 3, 4, 4)	12	85	128-elt discrete	Lean ✓
(1, 1, 1, 2, 2, 4, 4)	16	83	uniform magnitude	Lean ✓
(1, 1, 2, 2, 2, 3, 4)	16	83	phase-only + families	Lean ✓
(1, 1, 2, 2, 3, 3, 3)	16	85	continuous ($\sqrt{10}$ coeffs)	Lean ✓
(1, 1, 1, 2, 3, 3, 4)	16	85	isolated ($\sqrt{10}$ coeffs)	Lean ✓
(1, 1, 1, 2, 2, 3, 5)	14	99	isolated ($\sqrt{5}$ coeffs)	Lean ✓
(1, 1, 1, 2, 2, 2, 6)	10	115	proven infeasible	
(0, 1, 1, 2, 3, 3, 5)	14	78	proven infeasible	

$|S_0|$, non-auto from Lean #eval. 10 feasible, all 10 in Lean (exact arithmetic in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}, i)$).

Outlook

Outlook: Batch Verification of All 14,116 Codes

The distance-2 catalogue has 14,116 $((n, K, 2))$ codes. Currently: individual Lean files for representative examples.

Scaling strategy

1. Python exports each code's data as a `.lean` file (support, probabilities, angle vector)
2. Lean's `ExampleData.OK` bundle checks SS + screen + normalization + Z-KL
3. `native_decide` proves each in < 1 s
4. `lake build` compiles all 14K files

Already demonstrated

- `Ex522.lean`, `Ex622.lean`:
fully verified via `native_decide`
- Build time: 0.3–0.8 s each
- At 1 s/code: ~ 4 hours total
(embarrassingly parallel)

14,116 machine-checked proofs: within reach

Outlook: Analytical Family II & Beyond

Family II: 4-element support

$$C_0 = \{0^n, 1^n, e_1 \bar{e}_1, \bar{e}_1 e_1\}$$

Free parameter $\lambda \in [0, 1] \rightarrow$
continuous family.

Lean: `familyII_OK` $\forall (n, m, s, \lambda)$
passing screens.

λ -Families A–E (verified)

5 families at $n=5$, each covers ∞
values.

$$\Sigma_Z(\lambda) = \sum_i \langle Z_i \rangle^2 \in [\tfrac{1}{3}, 1]$$

What universality buys

- One 100-line file $\rightarrow$ **all** instantiations
- Proves **structure**, not cases
- Classification complete for $|C_0| \leq 4$

Open: analytical families for **distance 3?**

Needs universal proofs over $\mathbb{Q}(\sqrt{\cdot}, i)$ with
amplitude parameters — significantly
harder.

Outlook: Beyond BD_{16}

BD_{16} at $n=7$ is the **first step**. The framework extends naturally:

Other BD groups

$\text{BD}_{12}\text{--}\text{BD}_{36}$ ($m=6\text{--}18$)

Same pipeline; number field adapts to ω_m .

Zhang–Zeng: dozens of $((7, 2, 3))$ candidates

$n=8$: $T^{1/4}$

BD_{64} ($m=32$): transversal $\pi/16$.

Intersection test still effective.

Exceptional groups

$2I$ ($|G|=120$): 5th level

$2T$ ($|G|=24$)

$2O$ ($|G|=48$)

Two-layer architecture still applies.

Long-term vision: a formal library of quantum codes with **machine-verified** fault-tolerance properties — AI discovers, Lean certifies.

Summary

Summary

Distance 2

- Full SSLP pipeline in Lean 4
- Residue-shift screen $\Rightarrow d \geq 2$ automatic
- 14,116 codes certified
- Analytical families: one proof $\rightarrow \infty$ codes

Codebase

~4,300 lines Lean 4, 0 sorry
github.com/LionSR/lean-qec

Distance 3 (new)

- Intersection test filters most of 210 KL
- GPT-5.2 $\rightarrow$ closed-form amplitudes
- Exact arithmetic in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, i)$
- 8 $\text{BD}_{16}((7, 2, 3))$ codes: first machine-checked $d=3$ proofs

What's next

Batch 14K codes $\cdot \mathbb{Q}(\sqrt{5})$ extension
 $\cdot \text{BD}_{12} - \text{BD}_{36} \cdot n=8 \cdot 21/2T/20$

Thank you

arXiv:2510.20728 · github.com/LionSR/lean-qec